

What is VHDL?

- VHDL is a hardware description language .
- VHDL means **Very High Speed Integrated Circuit Hardware Description Language**
- You can do circuit descriptions with VHDL. Hardware description in digital circuit design
- You can design a gate (and , or) and a system with commands
- VHDL became one of industry's standard languages used to describe digital systems.
- Widely used in Industry
- You can describe and simulate complex digital systems

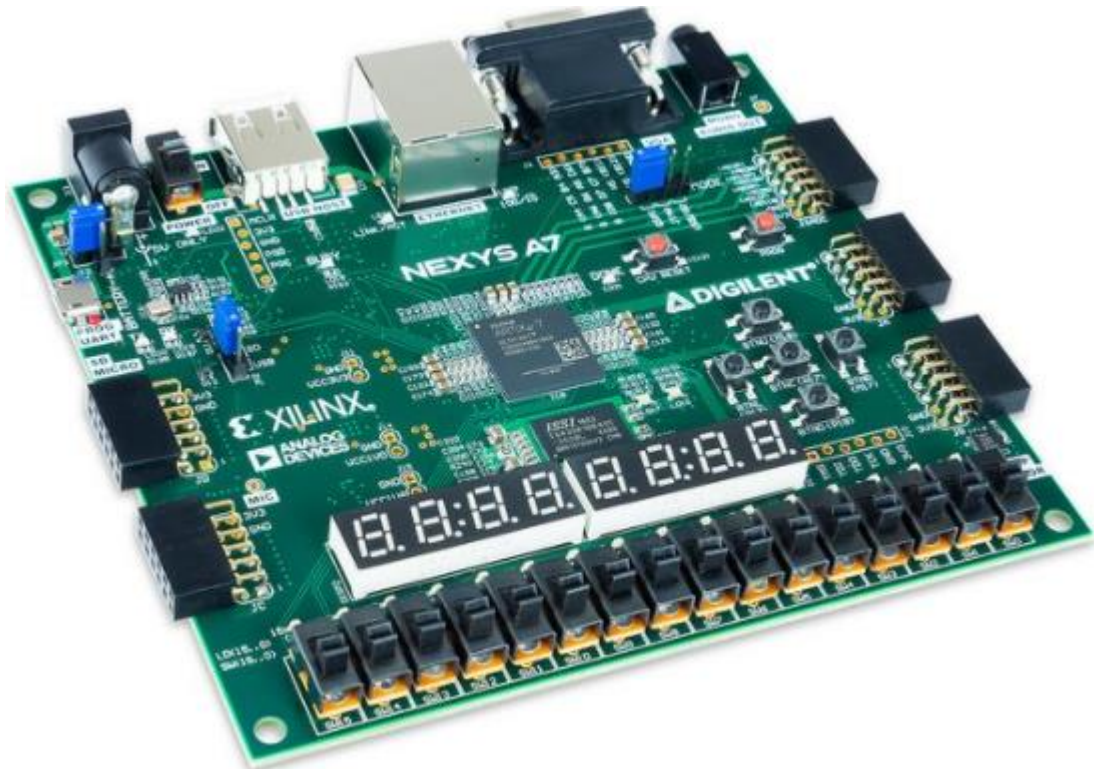
VHDL Languages

- Verilog
- ABEL
- XILINX

What is FPGA?

- Field programmable gate array
- This is an integrated circuit to be configured by a customer or designer





- The FPGA configuration is generally specified using a hardware description language (HDL)
- They contain a array of programmable logic blocks and a hierarchy of "reconfigurable interconnects" that allow the blocks to be "wired together", like many logic gates that can be inter-wired in different configurations
- Logic blocks can be configured to perform complex combinational functions or simple gates such as and , or.

Advantages of FPGA

- **Better Performance**

A general CPU is unable to perform parallel processing, giving FPGAs the upper hand as they can perform processing and calculation in parallel at a faster rate. Carefully designed FPGA can execute any function faster than a CPU which is running software code in a sequential fashion.

- **Programmability**

What is the greatest advantage that FPGAs have over any other alternative is the fact that they are reprogrammable. This means that even after the circuit has been designed and implemented, FPGAs can still be modified, updated, and completely change its functionality to perform a completely different task than before

Reprogrammability reduces the efforts and cost required for the long-term maintenance of these chips. You do not need to invest in replacing or redesigning new hardware when the old one becomes out of date — you can simply update its code and program it in the field with the new functionality

- **Cost Efficiency**

Since FPGAs can be reprogrammed again and again they prove to be extremely cost effective in the long run even though they may pose higher unit costs. They rid you of the need to cover for recurring any bug related costs that you may get stuck paying if you were to opt for an ASIC. ASICs also have heavy non-recurring expenses which are skipped altogether when it comes to FPGAs in addition to the fact that you do not require the use of any costly and expensive tools to design or configure your FPGA chip.

ASICs also prove to be more expensive in the long run as they must be completely redesigned along with hardware if updates and enhancements are needed. FPGAs do not share these costs as they can be reprogrammed for next to nothing.

- **Parallel Task Performance**

Chips that perform data processing in a sequential manner tend to not be used time critical applications. FPGAs can be designed to include multiple blocks processing data in parallel. This means that it can offer much greater scalability as compared to other processors such as ASICs and MCUs as well as time critical data processing.

- **Prototyping**

As mentioned before, FPGAs are reprogrammable and reusable. This makes them the perfect choice for prototyping purposes – especially for ASIC validation purposes. Before you tapeout your ASIC, it is important to determine whether your ASIC design is functioning and successfully achieves the purpose it has been designed for. ASICs are difficult and incredibly expensive to manufacture which means that if you end up with a chip that needs modification, you will have to invest a considerable amount of time and money to redesign the ASIC. With FPGAs, the reprogrammability feature allows you to perform test runs by manipulating the programming and determining the ideal configuration on one chip only. Once you have completed prototyping and determined what the best solution is, you can easily convert the FPGA into a permanent ASIC and deploy it for use.

- **Faster Time to Market**

One of the most significant advantages of FPGAs is that it allows you to finish the development of your product in a very short amount of time, meaning shorter time to market. FPGA design tools are easy to use and do not require a long learning curve.

In addition to that, FPGAs are designed in a higher description language called HDL which is also a modular programming code. Using HDL code, such as VHDL or Verilog makes the design process extremely fast and efficient.

FPGAs can very easily be reprogrammed at the software to test and validate the design in the lab. Modification can take place very quickly allow bug fixes in a short cycle time.

- **Simpler Design Cycles**

Compared to ASICs, FPGAs have simpler design cycles meaning the design tools take care of the major function by themselves including placement, routing, and timing in reference to the specifications you have set for it. As such, they require almost little to no manual intervention when converting the program code into a downloadable design.

If the design does not perform as expected, a new downloadable code can be prepared in a matter of hours – making the entire design cycle simple and quick.

- **Adaptability**

Because of their reprogrammability, FPGAs enable you to adapt and modify at the level of the customer rather than recreate the product whenever you need to make any updates. Even after the product has been finalized, developed, and delivered, you can update it and adapt it to the needs of the customer with minimum fuss.

- **Real Time Application**

Like mentioned before, FPGAs are perfect for time critical systems due to their more efficient processing architecture. As such, they are ideal for real time applications as they can perform more processing in a shorter period of time as compared to other alternative in the market.

- **System on Chip**

Since the 90's FPGAs have become larger in terms of gate count. This made it possible to include CPU cores inside the FPGA – from a single core to a multiple number of cores, alongside a custom hardware code. By combining CPU cores and hardware code in a single chip making the FPGAs to function as system on chip.

Disadvantages of FPGA

➡ The programming of FPGA requires knowledge of VHDL/Verilog programming languages as well as digital system fundamentals. The programming is not as simple as C programming used in processor based hardware. Moreover engineers need to learn use of simulation tools.

➡ The power consumption is more and programmers do not have any control on power optimization in FPGA. No such issues in ASIC.

➡ Once any particular FPGA is selected and used in the design, programmers need to make use of resources available on the FPGA IC. This will limit the design size and features. TO avoid such situation, appropriate FPGA need to be chosen at the beginning itself.

➡ FPGAs are better for prototyping and low quantity production. When the quantity of FPGAs to be manufactured increases, cost per product also increases. This is not the case with ASIC implementation.